

## Paul N Zivich

### Education

- 2016 – 2021 **Ph.D.**, *Department of Epidemiology*, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, NC  
Dissertation: Risk of influenza under realistic influenza vaccination interventions among university students
- 2014 – 2016 **M.P.H.**, *Department of Epidemiology*, College of Public Health, The Ohio State University, Columbus, OH
- 2011 – 2015 **B.S.P.H.**, College of Public Health, The Ohio State University  
Columbus, OH

### Experience

- 2023 – **Research Assistant Professor**, *Department of Epidemiology*, University of North Carolina at Chapel Hill, Chapel Hill, NC
- 2023 – **Participating Faculty**, *Biostatistics Core*, UNC Center for AIDS Research (CFAR), Chapel Hill, NC
- 2021 **Postdoctoral Research Associate**, *Causal Inference Research Laboratory*, University of North Carolina at Chapel Hill, Chapel Hill, NC
- 2015–2016 **Student Intern Researcher**, *Ohio Department of Health*, Columbus, OH
- 2015 **Epidemiology Intern**, *Cleveland Department of Public Health*, Cleveland, OH

### Grants

#### Current

- 2023 – 2028 **K01AI177102 NIH/NIAID (Zivich PN, PI)**, *Combining sources of information to improve HIV pre-exposure prophylaxis*, Principal Investigator  
Antiretroviral pre-exposure prophylaxis is an important tool for HIV prevention, with recent trials assessing new drug combinations. The limitations of clinical trials will increasingly require novel quantitative methods to reliably evaluate new interventions. The proposed study will use data from randomized trials, as well as novel methods, to address challenges in evaluation of new pre-exposure prophylaxis combinations.

#### Completed

- 2021 – 2023 **T32AI007001 NIH/NIAID (Adimora AA, PI)**, *Training in Sexually Transmitted Diseases and HIV*, Postdoctoral Trainee  
Provide training to doctoral students and postdoctoral fellows, including physicians, to create a cadre of young scientists with the skills in basic science, clinical medicine, epidemiology, and behavioral science to reduce the impact of sexually transmitted diseases, including HIV infection.

2018 – 2021 **T32HD091058 NIH/NICHD (Aiello AE & Hummer RA, MPI)**, *From Biological to Social Processes: Interdisciplinary Training in Life Course Research*, Predoctoral Trainee  
 Provide integrated social and biological/health sciences research training to junior scientists to better identify, understand, and produce new knowledge on the social and biological pathways to health across the life course, thereby identifying novel points of intervention to improve health and reduce health disparities.

## Publications

\* denotes student-led paper as primary mentor

### Peer Reviewed

1. Marthi A\*, Mottl AK, **Zivich PN**. "The (Mis)interpretation of Hazard Ratios in Clinical Trials" *Journal of the American Society of Nephrology* 2026; *In-Press*
2. Diepstra K, Westreich D, **Zivich PN**, Chapola J, Lemani C, Bula A, Lusewa C, Tang JH. "List randomization for prevalence estimation of sensitive behavioral data among women with HIV of reproductive age in Lilongwe, Malawi" *American Journal of Epidemiology* 2026; *In-Press*
3. Neuroth LM, Swilley-Martinez ME, **Zivich PN**, Edwards JK. "Pooled multinomial logistic regression for parametric g-computation in the presence of competing events" *Epidemiology* 2026; *Accepted*
4. Cole SR, **Zivich PN**, Shook-Sa BE, Richardson DB, Hudgens MG, Edwards JK. "Combining observational studies to reduce multiple biases" *Epidemiology* 2026; *Accepted*
5. Edwards JK, **Zivich PN**, Shook-Sa BE, Cole SR. "The Rogan-Gladen estimator for outcome misclassification" *American Journal of Epidemiology* 2026; *Accepted*
6. Baggett CD, Kuo T-M, **Zivich PN**, Rudolph JE, Ribis KM, Golden SD. "Assessing the causal effect of tobacco retail density on cardiovascular and pulmonary disease hospitalizations in the United States" *Preventive Medicine Reports* 2026; 66:103473
7. **Zivich PN**, Shook-Sa BE, Cole SR, Lofgren ET, Edwards JK. "Accounting for Missing Data in Public Health Research Using a Synthesis of Statistical and Mathematical Models" *Journal of Epidemiology and Community Health* 2026; 80(5):352-356.
8. **Zivich PN**, Klose M, DeMonte JB, Shook-Sa BE, Cole SR, Edwards JK. "An Improved Pooled Logistic Regression Implementation" *Epidemiology* 2026; 37(3):311-314.
9. Shook-Sa BE, Cole SR, **Zivich PN**, Edwards JK, Krajewski TJ, Feeney T, Adimora A, Hudgens MG. "A Primer on Large-Sample Statistical Inference for Epidemiologists" *Epidemiologic Methods* 2026; 15(1):20250036.
10. Edwards JK, Cole SR, **Zivich PN**, Ackerman B, Napravnik S, Henderson H, Lash T, Shook-Sa BE. "Risk functions with outcome measurement error" *Biostatistics* 2026; 27(1):kxaf052.
11. Wiener C, **Zivich PN**, Kurth T, Funk MJ, Breskin A, Berger K, Cole SR. "Causal identification conditions for the effect of treatment in the treated: Illustration using the Northwest Germany Stroke Registry" *Epidemiology* 2026; 37(1):57-66.
12. Edwards JK, Shook-Sa BE, Bakoyannis G, **Zivich PN**, Herce ME, Cole SR. "Accounting for Misclassification of Cause of Death in Weighted Cumulative Incidence Functions for Causal Analyses" *Statistics in Medicine* 2025; 44(23-24):e70281.
13. **Zivich PN**, Lu H. "Constructing g-computation estimators: two case studies in selection bias" *Epidemiology* 2025; 37(1):50-56.
14. Shook-Sa BE, **Zivich PN**, Cole SR, Rosenberg NE, Hudgens MG, Donnell DJ, Moyo S, Zuma K, Ayles H, Bock P, Eron JJ, Hayes RJ, Edwards JK. "Examining the effect of universal testing and treatment strategies for HIV prevention in Zambia and South Africa: generalizing the results of the HPTN 071 (PopART) trial" *Journal of the International AIDS Society* 2025; 28(12):e70062
15. Lu H, **Zivich PN**, Rudolph JE, Liew Z, Mukherjee B, Li F. "Revisiting Representativeness". *Interna-*

*tional Journal of Epidemiology* 2025; 54(4):dyaf109

16. **Zivich PN**, Naimi AI. "A Primer on Neural Networks". *American Journal of Epidemiology* 2025; 194(6):1473-1475.
17. **Zivich PN**, Cole SR, Edwards JK, Shook-Sa BE, Breskin A, Hudgens MG. "Bridged treatment comparisons: an illustrative application in HIV treatment". *American Journal of Epidemiology* 2025; 194(6):1687-1694.
18. Edwards JK, Breger TL, Cole SR, **Zivich PN**, Shook-Sa BE, Sadinski LM, Westreich D, Edmonds A, Ramirez C, Ofotokun I, Kassaye S, Brown T, Konkole-Parker D, Stosor V, Bolan R, Krier S, Jones DL, D'Souza G, Cohen M, Tien P, Taylor T, Anastos K, Drummond MB, Floris-Moore M. "Right censoring and mortality in the Multicenter AIDS Cohort Study and Women's Interagency HIV Study". *Epidemiology* 2025; 36(4):511-519.
19. Cole SR, Breskin A, Shook-Sa BE, **Zivich PN**, Hudgens MG, Edwards JK. "Five Facts About Influence Functions". *Epidemiology* 2025; 36(4):467-472.
20. Klose M, **Zivich PN**, Cole SR. "Revisiting the Population Attributable Fraction". *Epidemiology* 2025; 36(4):482-486.
21. **Zivich PN**, Shook-Sa BE. "Estimating marginal structural model parameters for time-fixed, binary actions with g-computation and estimating equations". *American Journal of Epidemiology* 2025; 194(5):1464-1466.
22. Shook-Sa BE, **Zivich PN**, Lee C, Xue K, Ross RK, Edwards JK, Stringer JSA, Cole SR. "Double Robust Variance Estimation with Parametric Working Models". *Biometrics* 2025; 81(2):ujaf054.
23. Cole SR, Shook-Sa BE, **Zivich PN**, Edwards JK. "A Minimal Nontrivial Example of Higher Order Evidence". *American Journal of Epidemiology* 2025; 194(4):886-888
24. **Zivich PN**, Edwards JK, Shook-Sa BE, Lofgren ET, Lessler J, Cole SR. "Synthesis estimators for transportability with positivity violations by a continuous covariate". *Journal of the Royal Statistical Society: Series A* 2025; 188(1):158-180.
25. **Zivich PN**, Ross RK, Shook-Sa BE, Cole SR, Edwards JK. "Empirical sandwich variance estimator for iterated conditional expectation g-computation". *Statistics in Medicine* 2024; 43(29): 5562-5572.
26. Edwards JK, Cole SR, **Zivich PN**, Hudgens MG, Breger TL, Shook-Sa BE. "Semiparametric g-computation for survival outcomes with time-fixed exposures: an illustration". *Annals of Epidemiology* 2024; 96:24-31.
27. Cole SR, Edwards JK, Breskin A, Rosin S, **Zivich PN**, Shook-Sa BE, Hudgens MG. "Rejoinder: Combining Information with Fusion Designs, and Roses by Other Names". *American Journal of Epidemiology* 2024; 193(5): 751-752.
28. Shi AX, **Zivich PN**, Chu HA. "Comprehensive Review and Tutorial on Confounding Adjustment Methods for Estimating Treatment Effects Using Observational Data". *Applied Sciences* 2024; 14, 3662.
29. Ross RK, **Zivich PN**, Stringer JSA, Cole SR. "M-estimation for common epidemiological measures: introduction and applied examples". *International Journal of Epidemiology* 2024; 53(2): dyae030.
30. **Zivich PN**, Cole SR, Edwards JK, Glidden DV, Das M, Shook-Sa BE, Shao Y, Mehrotra ML, Adimora AA, Eron JJ. "HIV prevention among men who have sex with men: tenofovir alafenamide combination preexposure prophylaxis versus placebo". *The Journal of Infectious Diseases* 2024; 229(4): 1123-1130.
31. Ross RK, Cole SR, Edwards JK, **Zivich PN**, Westreich D, Daniels JL, Price JT, Stringer JSA. "Leveraging external validation data: the challenges of transporting measurement error parameters". *Epidemiology* 2024; 35(2): 196-207.
32. Cole SR, Shook-Sa BE, **Zivich PN**, Edwards JK, Richardson DB, Hudgens MG. "Higher-order evidence". *European Journal of Epidemiology* 2024; 39(1): 1-11.
33. **Zivich PN**, Edwards JK, Lofgren ET, Cole SR, Shook-Sa BE, Lessler J. "Transportability without positivity: a synthesis of statistical and simulation modeling". *Epidemiology* 2024; 35(1): 23-31.

34. Shook-Sa BE, **Zivich PN**, Rosin SP, Edwards JK Adimora AA, Hudgens MG, Cole SR. "Fusing Trial Data for Treatment Comparisons: Single versus Multi-Span Bridging". *Statistics in Medicine* 2023; 43(4): 793-815.
35. Martinez RM, Andrabi N, Goodwin AN, Wilbur RE, Smith NR, **Zivich PN**. "Conceptualization, operationalization, and utilization of race and ethnicity in major epidemiology journals 1995-2018: a systematic review". *American Journal of Epidemiology* 2023; 192(3): 483-496.
36. Cole SR, Edwards JK, **Zivich PN**, Shook-Sa BE, Hudgens MG, Stringer JSA. "Reducing Bias in Estimates of Per Protocol Treatment Effects: A Secondary Analysis of a Randomized Clinical Trial". *JAMA Network Open* 2023; 6(7): e2325907
37. Cole SR, **Zivich PN**, Edwards JK, Shook-Sa BE, Hudgens MG. "Sensitivity Analyses for Means or Proportions with Missing Outcome Data". *Epidemiology* 2023; 34(5): 645-651.
38. **Zivich PN**, Cole SR, Edwards JK, Mulholland GE, Shook-Sa BE, Tchetgen Tchetgen E. "Introducing Proximal Causal Inference in Epidemiology". *American Journal of Epidemiology* 2023; 192(7): 1224-1227.
39. Renson A, Hudgen MG, Keil A, **Zivich PN**, Aiello AE. "Identifying and estimating effects of sustained interventions under parallel trends assumptions". *Biometrics* 2023; 79(4): 2998-3009.
40. Cole SR, Edwards JK, Breskin A, Rosin S, **Zivich PN**, Shook-Sa BE, Hudgens MG. "Illustration of Two Simple Fusion Designs and Estimators". *American Journal of Epidemiology* 2023; 192(3): 467-474.
41. Edwards JK, Cole SR, Shook-Sa BE, **Zivich PN**, Zhang N, Lesko CR. "When does differential outcome misclassification matter for estimating prevalence". *Epidemiology* 2023; 34(2):192-200.
42. Filiatreau LM, **Zivich PN**, Edwards JK, Mulholland GE, Max R, Westreich D. "Optimizing SARS-CoV-2 pooled testing strategies through differentiated pooling for distinct groups". *American Journal of Epidemiology* 2023; 192(2): 246-256.
43. Cole SR, **Zivich PN**, Edwards JK, Ross RK, Shook-Sa BE, Price JT, Stringer JSA. "Missing Outcome Data in Epidemiologic Studies". *American Journal of Epidemiology* 2023; 192(1):6-10.
44. **Zivich PN**, Hudgens MG, Brookhart MA, Moody J, Weber DJ, Aiello AE. "Targeted maximum likelihood estimation of causal effects with interference: a simulation study". *Statistics in Medicine* 2022; 41(23): 4554- 4577.
45. **Zivich PN**, Shook-Sa BE, Edwards JK, Westreich D, Cole SR. "On the use of covariate supersets for identification conditions". *Epidemiology* 2022; 33(4):559-562.
46. Noppert GA, Martin CL, **Zivich PN**, Aiello AE, Harris KM, O'Rand A. "Adolescent neighborhood disadvantage and memory performance in young adulthood" *Health & Place*. 2022; 75:102793
47. Smith MJ, Maringe C, Rachet B, Mansournia MA, **Zivich PN**, Cole SR, Luque-Fernandez MA. "Tutorial: Introduction to computational causal inference using reproducible Stata, R and Python code". *Statistics in Medicine*. 2022; 41(2):407-432
48. **Zivich PN**, Cole SR, Breskin A. "Twister plots for time-to-event studies". *American Journal of Epidemiology*. 2021; 190(12):2730-2731
49. **Zivich PN**, Volfovsky A, Moody J, Aiello AE. "Assortativity and Bias in Epidemiologic Studies of Contagious Outcomes: A Simulated Example in the Context of Vaccination". *American Journal of Epidemiology*. 2021; 190(11):2442-2452
50. Ciccone EJ, **Zivich PN**, Lodge EK, Zhu D, Law E, Taylor JL, Chung C, Xu J, Volfovsky A, Beatty C, Abernathy H, King E, Garrett HE, Markmann AJ, Rebuli ME, Sellers S, Weber DJ, Reyes R, Alvian N, Juliano JJ, Boyce RM, Aiello AE. "SARS-CoV-2 Infection in Healthcare Personnel and Their Household Contacts at a Tertiary Academic Medical Center: Protocol for a Longitudinal Cohort Study". *JMIR Research Protocols*. 2021; 10(4): e25410
51. **Zivich PN**, Breskin A. "Machine learning for causal inference: on the use of cross-fit estimators". *Epidemiology* 2021; 32(3):393-401
52. Hagaman A, LeMasters K, **Zivich PN**, Sikander S, Maselko J. "Longitudinal effects of perinatal

- social support on maternal depression: a marginal structural modeling approach". *Journal of Epidemiology and Community Health*. 2021; 75:936-943
53. LeMasters K, Bates LM, Chung EO, Gallis JA, Hagaman A, Scherer E, Sikander S, Staley BS, Zalla LC, **Zivich PN**, Maselko J. "Adverse Childhood Experiences and Depression Among Women in Rural Pakistan". *BMC Public Health*. 2021; 21(1):1-11
  54. **Zivich PN**, Huang W, Eisenberg MC, Walsh A, Aiello AE. "Measuring office workplace interactions and hand hygiene behaviors through electronic sensors: a feasibility study". *PLoS One*. 2021; 16(1):e0243358
  55. Smith NR, **Zivich PN**, Frerichs L, Moody J, Aiello AE. "A guide for choosing community detection algorithms in social network studies: the question-alignment approach". *American Journal of Preventative Medicine* 2021; 59(4):597-605
  56. Agler RA, **Zivich PN**, Kawende B, Behets F, Yotebieng M. "Postpartum depressive symptoms following implementation of the 10 steps to successful breastfeeding program in Kinshasa, Democratic Republic of Congo: a cohort study". *PLoS Med*. 2021; 18(1):e1003465
  57. **Zivich PN**, Eisenberg MC, Monto A, Uzicanin A, Baric RS, Sheahan T, Rainey J, Gao H, Aiello AE. "Transmission of viral pathogens in social network of university students: the eX-FLU study". *Epidemiology and Infection*. 2020; 148
  58. Aiello AE, Renson A, **Zivich PN**. "Social media- and internet-based surveillance for public health". *Annual Reviews of Public Health*. 2020; 41
  59. **Zivich PN**, Kiketa L, Kawende B, Lapika B, Behets F, Yotebieng M. "Effect of Family Planning Counseling After Delivery on Contraceptive Use at 24 Weeks Postpartum in Kinshasa, Democratic Republic of Congo". *Maternal and Child Health*. 2019; 23(4):530-537
  60. **Zivich PN**, Grabenstein JD, Becker-Dreps SI, Weber DJ. "Streptococcus pneumoniae Outbreaks and Implications for Transmission and Control: A Systematic Review". *Pneumonia* 2018; 10(11)
  61. **Zivich PN**, Yotebieng M, Lapika B, Behets F. "Implementation of Ten Steps to successful Breastfeeding and incidence of diarrhea, respiratory tract infection, and hospitalization among 0-6 month infants in Kinshasa, Democratic Republic of Congo: A randomized controlled trial". *Maternal and Child Health*. 2018; 22(5):762-771
  62. **Zivich PN**, Gancz A, Aiello AE. "Effect of Hand Hygiene in the Office Workplace on Infectious Diseases: A Systematic Review" *American Journal of Infection Control*. 2018; 46(4):448-455
  63. **Zivich PN**, Tatham L, Lung K, Tien J, Bollinger CE, Bower JK. "The association of influenza vaccine status and timing with outcomes among patients in Columbus, Ohio (2012-2015)". *Epidemiology and Infection*. 2017; 145(15):3284-3293
  64. **Zivich PN**, Kiketa L, Kawende B, Lapika B, Yotebieng M. "Vaccination coverage and timelines among children 0-6 months, Democratic Republic of Congo: a prospective cohort study". *Maternal and Child Health*. 2017; 21(5):1055-1064

## Letters, Commentaries, & Perspectives

1. Feeney T, **Zivich PN**. "Causal inference is hard, and advanced statistical analysis is not enough" *The BMJ* 2025; 391:r2618
2. Martinez RAM, Smith NR, Wilbur RE, Andrabi N, Goodwin AN, **Zivich PN**. "Not as we are, but as we could be: the role of agency and structure towards the improvement of race, ethnicity, and racism in epidemiology". *American Journal of Epidemiology* 2025; 194(4):936–940.
3. Lu H, Howe CJ, **Zivich PN**, Gonsalves GS, Westreich D. "The Evolution of Selection Bias in the Recent Epidemiologic Literature – A Selective Overview". *American Journal of Epidemiology* 2025; 194(3):580-584.
4. **Zivich PN**. "RE: 'Estimating the effect of a treatment when there is Nonadherence in a trial'". *American Journal of Epidemiology* 2025; 194(2):552-553.

5. Li CX\*, **Zivich PN**. "Invited Commentary: Mixing multiple imputation and bootstrapping for variance estimation". *American Journal of Epidemiology* 2024; 193(10):1477-1481.
6. **Zivich PN**. "Commentary: The Seedy Side of Causal Effect Estimation with Machine Learning". *Epidemiology* 2024; 35(6):787-790.
7. Cole SR, **Zivich PN**, Edwards JK, Ross RK, Shook-Sa BE. "Five Authors Reply". *American Journal of Epidemiology* 2024; 193(3):562.
8. Martinez RAM, Smith NR, Wilbur RE, Andrabi N, Goodwin AN, **Zivich PN**. "Threats to Equity: Missing Methodological Details about Race and Ethnicity in Health Research". *Health Affairs Forefront* March 2023
9. **Zivich PN**, Ross RK. "RE: 'Using numerical methods to design simulations: revisiting the balancing intercept'". *American Journal of Epidemiology*. 2022; 191(10), 1833-1834.
10. Smith NR, **Zivich PN**, Frerichs L. "Social influences on obesity: current knowledge, emerging methods, and directions for future research and practice". *Current Nutrition Reports*. 2020; 9(1):31-41

## Book Chapters

1. **Zivich PN**, Breskin A, Kennedy EH. "Machine learning and causal inference". *WileyStatsRef: Statistics Reference Online*. New York: Wiley & Sons.

## Non-Peer-Reviewed Publications

1. Naimi AI, Nabi R, Collin LJ, **Zivich PN**, Cole SR. "A Law of Iterated Expectation Primer for Causal Inference" *arXiv:2606.20078*
2. D'Alessio AS\*, Neuroth LM, Edwards JK, Martin CL, **Zivich PN**. "Novel g-computation algorithms for time-varying actions with recurrent and semi-competing events" *arXiv:2603.10169*
3. **Zivich PN**. "Unifying Statistical and Mathematical Modeling with Causal Inference" *arXiv:2511.01960*
4. Hughes-Noehrer L, Parkes MJ, Stewart A, Wilson AJ, Collins GS, Riley RD, Mathur M, Fox MP, Islam N, **Zivich PN**, Feeney TJ. "Code Sharing in Healthcare Research: A Practical Guide and Recommendations for Good Practice" *arXiv:2510.19279*
5. **Zivich PN**, Cole SR, Greifer N, Montoya LM, Kosorok MR, Edwards JK. "Confidence Regions for Multiple Outcomes, Effect Modifiers, and Other Multiple Comparisons" *arXiv:2510.07076*
6. **Zivich PN**, Cole SR, Shook-Sa BE, DeMonte JB, Edwards JK. "Estimating equations for survival analysis with pooled logistic regression" *arXiv:2504.13291*
7. Shahn Z, **Zivich PN**, Renson A. "Structural Nested Mean Models Under Parallel Trends with Interference" *arXiv:2405.11781*
8. **Zivich PN**, Cole SR, Westreich D. "Positivity: Identifiability and Estimability". *arXiv:2207.05010*
9. **Zivich PN**, Klose M, Cole SR, Edwards JK, Shook-Sa BE. "Delicatessen: M-Estimation in Python". *arXiv:2203.11300*
10. Martinez RAM, Wilbur RE, Andrabi N, Goodwin AN, Smith NR, **Zivich PN**. "Conceptualization, operationalization, and utilization of race and ethnicity in major medical journals 1995-2018: a systematic review". *medRxiv:2022.03.07.22271661*

## Invited Presentations

1. **Zivich PN**. (2026) "Causal Parameters for Public Health Policy". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 24-26, 2026. Phoenix, AZ, USA.
2. **Zivich PN**. (2026) "Sensitivity analyses for positivity violations". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 24-26, 2026. Phoenix, AZ, USA.
3. **Zivich PN**. (2026) "Synthesizing Statistical and Mathematical Models to Address Non-Positivity".

Oral Presentation at the American Causal Inference Conference (ACIC). May 11-14, 2026. Salt Lake City, UT, USA.

4. **Zivich PN.** (2026) "Fusion Designs, Fusion Estimators, Fusion Diagnostics and Performance". Oral Presentation at the National Institutes of Health (NIH) HIV Vertical Transmission Prevention Workshop. May 11-12, 2026. Virtual.
5. **Zivich PN.** (2026) "Improving Comparisons Across Studies: the Case of Long-acting Injectable Cabotegravir versus Placebo for HIV Prevention". Oral Presentation at the 28<sup>th</sup> International Workshop on HIV and Hepatitis Observational Databases. March 26-28, 2026. Thessaloniki, Central Macedonia, GRC.
6. **Zivich PN.** (2025) "Representativeness (or who does this estimate apply to?)". Oral Presentation at the 11<sup>th</sup> Annual University of North Carolina at Chapel Hill Pharmacoepidemiology Methods Symposium. September 11, 2025. Virtual.
7. **Zivich PN.** (2025) "Stabilizing Causal Estimates: Tackling Random Seed Dependence in Machine Learning". Oral Presentation at the International Conference on Statistics and Data Science (ICSIDS). June 23-25, 2025. Vancouver, BC, CAN.
8. **Zivich PN.** (2025) "Using observational data to improve inference in randomized trials" Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 10-13, 2025. Boston, MA, USA.
9. **Zivich PN.** (2025) "Estimating the risk of influenza under differing distributions of vaccination among university students". Oral Presentation at the American Causal Inference Conference (ACIC). May 13-16, 2025. Detroit, MI, USA.
10. **Zivich PN,** Edmonds A, Floris-Moore M, Ramirez C, Pence BW, Alcaide ML, Jones DL, Kassaye S, Konkole-Parker D, Sheth AN, Topper EF, Rana A, Westreich D. (2025) "Depressive Symptoms on the Incidence of Sexual Transmitted Infections Among Reproductive-Age Women". Poster Presentation at Conference on Retroviruses and Opportunistic Infections (CROI). March 9-12, 2025. San Francisco, CA, USA.
11. **Zivich PN.** (2025) "Addressing Positivity Violations Using a Synthesis of Statistical and Mathematical Models". Oral Presentation at Nationwide Children's Hospital. February 25, 2025. Columbus, OH, USA.
12. **Zivich PN.** (2025) "An Introduction to Estimating Equations". Oral Presentation at The Ohio State University. February 24, 2025. Columbus, OH, USA.
13. **Zivich PN.** (2025) "Effects of Influenza Vaccination Among University Students Using Targeted Maximum Likelihood for Interference". Oral Presentation for the ID Epi Spring Seminar Series hosted by the Center for Communicable Disease Dynamics at Harvard University. January 30, 2025. Cambridge, MA, USA.
14. **Zivich PN.** (2024) "Addressing Positivity Violations Using a Synthesis of Statistical and Mathematical Models". Oral Presentation at University of Nebraska Medical Center. November 5, 2024. Omaha, NE, USA.
15. **Zivich PN.** (2024) "Transportability Without Positivity: A Synthesis of Statistical and Simulation Modeling". Lilienfeld Postdoctoral Prize Paper Award at the 57th Society for Epidemiologic Research (SER) annual meeting. June 18-21, 2024. Austin, TX, USA.
16. **Zivich PN** (2024) "Addressing nonpositivity using statistical and mathematical models". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 18-21, 2024. Austin, TX, USA.
17. **Zivich PN.** (2024) "Transporting Average Causal Effects with Positivity Violations". Poster Presentation at the American Causal Inference Conference (ACIC). May 15-17, 2024. Seattle, WA, USA.
18. **Zivich PN.** (2023) "Extending Inferences to a Target Population Without Positivity". Oral Presentation at Extending Inferences to a New Target Population hosted by The Institute for Computational

and Experimental Research in Mathematics (ICERM) at Brown University. November 17-19, 2023. Providence, RI, USA.

19. **Zivich PN.** (2023) "Principled comparisons of treatment arms from different randomized trials in HIV research". Oral Presentation at the Joint Statistical Meetings. August 5-10, 2023. Toronto, Ontario, CA, USA.
20. **Zivich PN.** (2022) "Targeted maximum likelihood estimation with network-dependent data". Oral Presentation at the International Conference on Advances in Interdisciplinary Statistics and Combinatorics. October 7-9, 2022. Greensboro, NC, USA.
21. **Zivich PN.** (2022) "M-estimation". Oral Presentation at Centre for Statistical Methodology at the London School of Hygiene and Tropical Medicine. October 5, 2022. Virtual.
22. **Zivich PN.** (2022) "Why I use Python". Oral Presentation at the 43rd Annual Conference of the International Society for Clinical Biostatistics. August 21-25, 2022. Newcastle upon Tyne, UK.
23. **Zivich PN.** (2022) "M-estimation for fusion designs". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 14-17, 2022. Chicago, IL, USA.
24. **Zivich PN.** (2022) "Realistic policies on influenza vaccination and risk of subsequent influenza among university students: application of targeted maximum likelihood estimation with interference". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 14-17, 2022. Chicago, IL, USA.
25. **Zivich PN.** (2022) "Fusion to address systematic errors across data sources". Oral Presentation at the Emerging Challenges for Statistics and Data Sciences: Complex Data with Missingness, Measurement Errors, and High Dimensionality. May 23-27, 2022. University of British Columbia - Okanagan, Kelowna, BC, CAN.
26. **Zivich PN.** (2022) "Targeted Maximum Likelihood Estimation for Causal Inference with Network Data". Oral Presentation at Duke Social Networks and Health Workshop. May 10-12, 2022. Durham, NC, USA.
27. **Zivich PN.** (2021) "Efficient machine learning for causal effects". Health Data Science for Impact Workshop at Singapore Institute for Clinical Sciences. March 16-18, 2021. Virtual.
28. **Zivich PN, Hudgens MG, Moody J, Brookhart MA, Weber DJ, Aiello AE.** (2020) "Examining the utility of targeted maximum likelihood estimation for network data". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. December 15-18, 2020. Virtual.
29. **Zivich PN, Volfovsky A, Moody J, Aiello AE.** (2019) "Assortativity by Treatment in Social Networks and Bias in Population Health Studies". Oral Presentation at the Interdisciplinary Association of Population Health Scientists (IAPHs) Meeting. October 1-4, 2019. Seattle, WA, USA.
30. **Zivich PN, Edwards JK, Keil AP.** (2019) "An Inverse Probability Weighting Approach to Correct for Left Truncation". Oral Presentation at the Society for Epidemiologic Research (SER) annual meeting. June 18-21, 2019. Minneapolis, MN, USA.
31. **Zivich PN, Volfovsky A, Moody J, Aiello AE.** (2019) "Assortativity by Treatment in Social Networks and Bias". Poster Presentation at the Symposium on Biosocial Approaches to Population Health across the Life Course. May 13, 2019. Chapel Hill, NC, USA.
32. **Zivich PN.** (2018) "Better understanding transmission: a social network approach". Invited oral presentation at University of North Carolina at Chapel Hill Social Epidemiology Seminar. September 26, 2018. Chapel Hill, NC, USA.
33. **Zivich PN, Volfovsky A, Moody J, Aiello AE.** (2018) "Using Networks to better estimate Vaccine Effectiveness". Oral Presentation at Social Networks and Health. May 14, 2018. Durham, NC, USA.
34. **Zivich PN, Eisenberg MC, Walsh A, Aiello AE.** (2017) "Smartphone as a Tool for Deep Epidemiologic Data on Social Networks". Oral Presentation in Infectious Disease session at the Society for Epidemiologic Research (SER) annual meeting. June 20-23, 2017. Seattle, WA, USA.
35. **Zivich PN, Tatham L, Lung K, Tien J, Bollinger CE, Bower JK.** (2016) "Influenza vaccination status



and outcomes among hospitalized patients in Ohio 2011-2014". Oral Presentation at the 30th Annual Hayes Graduate Research Forum. February 22, 2016. Columbus, OH, USA.

36. **Zivich PN**, Bower JK, Klein EG. (2015) "Perceptions of Tobacco Policies from National Social Fraternities and Sororities". Poster Presentation at the 20th Annual Denman Undergraduate Research Forum. March 25, 2015. Columbus, OH, USA.

## Teaching

- 2024 – **Co-Instructor**, *Advanced Epidemiologic Methods*, University of North Carolina at Chapel Hill  
Topics: causal survival analysis, inverse probability weighting, g-computation, joint effects, time-varying confounding, survival regression models
- 2023 – 2026 **Workshop Instructor**, *ABC's of M-estimation*, Society for Epidemiologic Research (SER) Annual Meeting
- 2026 **Participant**, *4<sup>th</sup> International Meeting on Teaching Epidemiology*, Epidemiology, Biostatistics and Prevention Institute (EBPI) at the Universität Zürich
- 2019 **Graduate Teaching Assistant**, *Epidemiologic Analysis of Time-to-Event Data*, University of North Carolina at Chapel Hill

## Guest Lectures

- 2026 **University of Houston**, *PHCA 7180: Seminar in Pharmaceutical Health Outcomes and Policy*  
Topics: g-methods for repeated measures and time-varying confounding
- 2026 **University of Nebraska Medical Center**, *EPI 998: Doctoral Special Topics: Advanced Epidemiology Theory & Methods*  
Topics: g-formula with time-to-event data
- 2025 **Yale University**, *CDE 634: Advanced Applied Analytic Methods in Epidemiology and Public Health*  
Topics: machine learning and artificial intelligence in epidemiology
- 2022 – 2023 **University of North Carolina at Chapel Hill**, *EPID 722: Epidemiologic Analysis of Time-to-Event Data*  
Topics: informative censoring, g-formula for time-to-event data

## Mentorship

### Chair or Co-Chair of Doctoral Committee

- Current Mark Klose, *Modern Approaches to Immortal Time Bias with Competing Events*, Chair: Paul Zivich, Department: Epidemiology
- Current Alena Sorensen D'Alessio, *Longitudinal cohort fusion methodology for life course epidemiology*, Co-Chairs: Paul Zivich, Chantel Martin, Department: Epidemiology

### Member of Doctoral Committee

- Current Lucas Neuroth, *Using a multi-state modeling approach to holistically assess progression through New Jersey's graduated driver license system*, Chair: Steve Marshall, Department: Epidemiology

- May 2026 Chelsea Martin, *Implementing A Longitudinal Study Design in Baseball: Injury and Functional Outcomes, Causal Approaches, and Knowledge User Perspectives to Improve Pitcher Health*, Chair: Stephen Marshall, Department: Epidemiology
- Dec 2025 Justin DeMonte, *Sequential trial emulation for observational vaccine effectiveness studies*, Chair: Michael Hudgens, Department: Biostatistics
- Dec 2025 Tianhao Song, *Triggering-Effect Models for Multi-Type Recurrent Events*, Chair: Anastasia Ivanova, Department: Biostatistics
- May 2025 Chanhwa Lee, *Causal inference under clustered interference and nonparametric efficient estimation*, Chair: Michael Hudgens, Department: Biostatistics

## Editorial Responsibilities

- 2026 – *Epidemiology*, Associate Editor
- 2026 – *Journal of the Royal Statistical Society, Series A*, Associate Editor
- 2024 – *Epidemiologic Methods*, Associate Editor

## Society & Conference Contributions

### Session Chair

- 2026 Society for Epidemiologic Research, *"Innovations in Sensitivity Analysis"*, Chairs: Paul Zivich, Stephen Cole  
June 24-26. Phoenix, AZ
- 2026 Society for Epidemiologic Research, *"Causal inference methods to learn about hypothetical interventions in the policy environment"*, Chairs: Paul Zivich, Alena Sorensen D'Alessio  
June 24-26. Phoenix, AZ
- 2026 American Causal Inference Conference, *"Some Positive Developments on the Positivity Assumption"*, Chair: Paul Zivich  
May 11-14. Salt Lake City, UT
- 2025 The Joint Statistical Meeting, *"Modern Statistical Methods in HIV Research"*, Chairs: Paul Zivich, Bonnie Shook-Sa, Michael Hudgens  
August 3-7. Nashville, TN
- 2025 Society for Epidemiologic Research, *"Leveraging Multiple Data Sources to Improve Epidemiology"*, Chairs: Paul Zivich, Stephen Cole  
June 11-13. Boston, MA
- 2024 American Causal Inference Conference, *"How certain should we be? Modern methods for sensitivity analysis"*, Chair: Paul Zivich  
May 15-17. Seattle, WA
- 2023 Society for Epidemiologic Research, *"Does Machine Learning Work? Designing Better Simulation Studies for Inference"*, Chairs: Paul Zivich, Ashley Naimi  
June 13-16. Portland, OR
- 2023 Society for Epidemiologic Research, *"Answering questions by combining data sources"*, Chairs: Paul Zivich, Jessie Edwards  
June 13-16. Portland, OR

- 2022 Society for Epidemiologic Research, *"The New Epidemiology of Fusion Designs"*,  
Chairs: Paul Zivich, Stephen Cole  
June 14-17. Chicago, IL

## **Mentorship**

- 2026 Society for Epidemiologic Research, *Mid-Year Dissertation Workshop*, Faculty Mentor  
2025 Society for Epidemiologic Research, *SERvisits Mentoring Program*, Faculty Mentor

## **Grant Reviews**

### **Ad hoc Reviews**

- 2026 Dutch Research Council (NWO)  
Knowledge and Innovation Covenant (KIC)  
2025 United States Centers for Disease Control and Prevention (CDC)  
Special Emphasis Panel: Post Exposure Prophylaxis for HIV and STI Prevention

## **Reviewer Contributions**

American Journal of Epidemiology, Annals of Applied Statistics, Annals of Epidemiology, Annals of Internal Medicine, Biometrics, Biostatistics, BMJ Open, Communications in Statistics – Case Studies and Data Analysis, Data Technologies & Applications, Epidemiology, Influenza and Other Respiratory Viruses, Informatics for Health and Social Care, International Journal of Epidemiology, International Journal of Obesity, Journal of Causal Inference, Journal of Clinical Epidemiology, Journal of Infectious Diseases, Journal of the Royal Statistical Society Series A, Lifetime Data Analysis, Nature – Scientific Reports, PLoS Computational Biology, PLoS Global Public Health, PLoS One, Pharmaceutical Statistics, Public Health Reviews, Social Forces, Social Networks, Statistics in Medicine, Vaccine

## **Honors and Awards**

- 2026 Gillings School Teaching Innovation Award, *Gillings School of Global Public Health, University of North Carolina at Chapel Hill*  
2025 Exemplary Reviewer, *Epidemiology*  
"Thanks to Our Reviewers 2025" *Epidemiology* 2026; 37(3):279-280  
2024 Lilienfeld Postdoctoral Prize Paper Award, *Society for Epidemiologic Research*  
Zivich PN et al. "Transportability without positivity: a synthesis of statistical and simulation modeling". *Epidemiology* 2024; 35(1): 23-31.  
2023 Research featured in 'Early Career Researchers: A Joint Collection from the *American Journal of Epidemiology* and the *European Journal of Public Health*'  
- Zivich PN et al. "Introducing Proximal Causal Inference in Epidemiology". *American Journal of Epidemiology* 2023; 192(7): 1224-1227.  
- Filiatreau LM et al. "Optimizing SARS-CoV-2 pooled testing strategies through differentiated pooling for distinct groups". *American Journal of Epidemiology* 2023; 192(2): 246-256.  
2017 – 2019 Social Network and Health Fellow at Duke Network Analysis Center  
2017 Member of Delta Omega, Honorary Society in Public Health  
2015 Sigma Xi Scientific Research Society

## ———— Open-Source Software Contributions

Python Delicatessen: creator, maintainer

zEpid: creator, maintainer

MossSpider: creator, maintainer

Lifelines: contributor